

CONSTRUCTION TECHNOLOGY

Construction technology has always played a central role in society. It is a multifaceted industry in which projects can result in products, systems, processes, or services and encompass various aspects of production, repair, and maintenance. In construction technology courses, students will learn about designing, constructing, and maintaining a variety of buildings and structures, and will gain experience with the tools, equipment, and processes commonly used in the field. Students will also learn about health and safety standards in the construction industry, building codes and regulations, and employment opportunities and careers in the various sectors of the industry, including electricity, carpentry, masonry, heating and cooling, and plumbing.

The construction technology courses initially focus on residential and light construction systems related to residential buildings and structures, and progress to more advanced residential and light commercial construction. The woodworking courses allow students to concentrate on furniture building or cabinet making.

The list of approved emphasis areas for construction technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Custom Woodworking, Grade 12

Workplace Preparation

TWJ4E

This course enables students to further develop knowledge and skills related to the planning, design, and construction of residential and/or commercial cabinets and furniture. Students will gain further experience in the safe use of common woodworking materials, tools, equipment, finishes, and hardware, and will learn about the entrepreneurial skills needed to establish and operate a custom woodworking business. Students will also expand their awareness of health and safety issues and environmental and societal issues related to woodworking, and will explore career opportunities that may be pursued directly after graduation.

Prerequisite: Custom Woodworking, Grade 11, Workplace Preparation

A. CUSTOM WOODWORKING FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** describe the features and applications of materials and processes used in custom woodworking;
- A2.** demonstrate an understanding and safe use of tools, equipment, and techniques for custom woodworking;
- A3.** use correct terminology to describe woodworking materials, tools, equipment, processes, and related concepts.

SPECIFIC EXPECTATIONS

A1. Materials and Processes

By the end of this course, students will:

- A1.1** describe the factors that affect the quality and classification of wood (*e.g., harvesting method, cutting, drying, flaws*);
- A1.2** describe the factors that should be considered when selecting and purchasing wood products (*e.g., cost, type of project, quality, flaws*);
- A1.3** describe various moulding profiles and their applications (*e.g., baseboard, casing, crown moulding*);
- A1.4** describe the advantages and drawbacks of manufactured materials used in custom woodworking (*e.g., plywood, particle board, oriented-strand board, plastics, steel, adhesives*);
- A1.5** compare the properties (*e.g., strength, stability, appearance*) of natural and manufactured materials that are used for the same function (*e.g., natural wood versus plywood or particleboard, linseed oil versus polyurethane*);
- A1.6** demonstrate an understanding of structural and non-structural components of woodworking projects (*e.g., structural: rails, stiles, corner braces; non-structural: trim, finishes, hardware, lazy Susan*);
- A1.7** identify and describe various processes for fabricating components of a woodworking project (*e.g., jointing, laminating, turning, milling*);
- A1.8** demonstrate a thorough understanding of various materials and processes used to finish

woodworking projects (*e.g., materials: sandpaper, stains, oils, varnishes; processes: sanding, detailing, brushing, spraying*);

- A1.9** describe procedures for mass-producing a product (*e.g., use of patterns, jigs, automated machinery, and/or modular construction*), and describe the advantages and disadvantages of mass production as opposed to custom woodworking (*e.g., quality, cost per unit*).

A2. Tools, Equipment, and Techniques

By the end of this course, students will:

- A2.1** demonstrate proficiency in using, maintaining, adjusting, and storing construction tools and equipment safely (*e.g., chisels, planes, measuring instruments, table saws, drills, lathes*);
- A2.2** demonstrate proficiency with commonly used measuring, layout, and assembly methods (*e.g., layout and measuring: determining and marking circumference, diameter, radius, angles, and rounded corners; assembly: fasteners, glues, milled joints*);
- A2.3** demonstrate proficiency in the selection and safe application of appropriate clamps, fasteners, and adhesives;
- A2.4** demonstrate proficiency in safely and accurately constructing and fitting commonly used joints (*e.g., butt, half lap, mortise and tenon, dovetail, dowel, mitre, rabbet, tongue and groove*).

A3. Terminology

By the end of this course, students will:

- A3.1** use correct terminology to identify and describe woodworking materials, tools, equipment, and processes (*e.g., materials: adhesives, wood types, fasteners; tools: router, mitre saw, biscuit jointer, T-bevel, brad nailer; equipment: dust-collection system, spray painting booth, air compressor; processes: levelling, squaring, making plumb, joining*);
- A3.2** use correct terminology to identify and describe concepts related to custom woodworking projects (*e.g., aesthetics, ergonomics, environmental stewardship, sustainability*);
- A3.3** use correct terminology to describe the distinguishing features of various cabinet and furniture styles (*e.g., gothic, baroque, early colonial*).

B. DESIGN, LAYOUT, AND PLANNING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** apply a design process and other problem-solving processes and techniques when planning a variety of woodworking projects, taking into account design principles, safety standards, and other relevant factors;
- B2.** prepare and interpret sketches and/or working drawings for a variety of woodworking projects;
- B3.** plan efficient and effective fabrication and assembly of residential and/or commercial woodworking projects;
- B4.** apply the mathematical skills required in the design and construction of woodworking projects.

SPECIFIC EXPECTATIONS

B1. Design

By the end of this course, students will:

- B1.1** explain the steps of a design process (see pp. 22–23), and apply them to plan and develop a variety of woodworking projects (*e.g., kitchen counter, cabinet, entertainment centre, coffee table, desk*);
- B1.2** use appropriate problem-solving processes and techniques (see pp. 21–23) to address challenges that arise in the course of various woodworking projects;
- B1.3** demonstrate how the overall design of a project is affected by design concepts and other factors (*e.g., structure, ergonomics, aesthetics, cost and availability of materials, customer needs*);
- B1.4** describe how to enhance the appearance and functionality of woodworking projects by using appropriate design elements (*e.g., line, space, texture, colour*) and principles (*e.g., balance, scale, proportion, emphasis, unity, continuity*);
- B1.5** explain the reasons for choosing particular materials for various woodworking projects (*e.g., strength, cost, appearance, ease of use, availability, environmental considerations*);
- B1.6** apply anthropometric measurements and ergonomic principles to projects (*e.g., for the height and width of a chair seat or a desk*);
- B1.7** describe the building codes and regulations that apply to built-in woodworking projects (*e.g., Ontario Building Code [Part 7 – Plumbing*

and Part 9 – Housing and Small Buildings], fire codes, municipal by-laws);

- B1.8** describe product standards that apply to woodworking projects (*e.g., Canadian Standards Association [CSA], Underwriters Laboratories of Canada [ULC], and National Lumber Grades Authority standards*).

B2. Working Drawings

By the end of this course, students will:

- B2.1** prepare and/or modify sketches and working drawings (*e.g., orthographic view, plan, elevation, section, detail drawing*), using manual and/or computer-assisted methods and appropriate metric and/or imperial units;
- B2.2** interpret a variety of working drawings accurately to produce woodworking projects (*e.g., determine dimensions, materials, and hardware placement*).

B3. Planning and Preparation

By the end of this course, students will:

- B3.1** plan and describe the interrelationship of structural and decorative elements in woodworking projects (*e.g., choose structural and finish materials that will bond to each other; choose hardware that suits the shape and style of the design*);

- B3.2** create a materials list itemizing all materials and components for a woodworking project;
- B3.3** plan and prepare for installation of various components used in woodworking projects (*e.g., make a work schedule and list of tools required, check stock on hand, order materials*);
- B3.4** plan an efficient sequence for fabricating components for a project (*e.g., use appropriate tools and techniques, do operations such as cutting and drilling in batches to reduce set-up time*);
- B3.5** lay out projects using a variety of measurement tools and equipment (*e.g., laser level, tape measure, ruler, scale, protractor*);
- B3.6** describe and, where appropriate, make modifications to woodworking projects to improve the quality and value of the project (*e.g., choose different materials, hardware, finishes, or methods of construction and assembly*).

B4. Mathematical Skills

By the end of this course, students will:

- B4.1** calculate dimensions for woodworking projects in appropriate metric and/or imperial units;
- B4.2** convert between fractions and decimals and between imperial and metric units, using appropriate charts, tables, software, and/or online conversion tools, to determine dimensions and quantities for woodworking projects (*e.g., convert lengths from inches to centimetres, areas from square metres to square feet, and volumes from gallons to litres*);
- B4.3** apply mathematical concepts and formulas (*e.g., Pythagorean theorem, volume and area formulas*) when preparing components of a woodworking project;
- B4.4** determine lengths, diameters, and quantities of fastening devices needed to assemble various woodworking projects (*e.g., lengths and gauges of screws, nails, and staples; diameters of dowels*);
- B4.5** prepare detailed, accurate estimates of quantities and costs of materials for woodworking projects (*e.g., lumber, hardware, finishes*), and of labour costs for these projects.

C. FABRICATION, ASSEMBLY, AND FINISHING SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** fabricate and assemble residential and/or commercial custom woodworking projects safely, accurately, and efficiently;
- C2.** prepare surfaces and apply finishing products, trim, and hardware correctly and safely.

SPECIFIC EXPECTATIONS

C1. Fabrication and Assembly

By the end of this course, students will:

- C1.1** apply techniques for using tools and materials safely and efficiently to reduce the cost of producing components that meet the required specifications (*e.g., follow proper procedures for operating machine tools; use stops, guides, or jigs when making sets of identical parts*);
- C1.2** use tools, equipment, and techniques to safely and accurately prepare project materials (*e.g., dress raw lumber; measure, cut, square, and drill stock*);
- C1.3** program a computer numerically controlled (CNC) machine to fabricate components for a woodworking project;
- C1.4** develop and apply appropriate quality-control measures to ensure precise dimensions and correct assembly (*e.g., accurate measurements and cuts, clean joints, true edges*);
- C1.5** install various components of a woodworking project in accordance with applicable codes, regulations, and standards.

C2. Finishing

By the end of this course, students will:

- C2.1** demonstrate a thorough understanding of how to prepare surfaces for finishes given the type of material, the desired finish, and the intended use of the project;
- C2.2** use appropriate tools, equipment, and techniques correctly and safely to install trim and hardware (*e.g., baseboards, hinges, pulls, casters*);
- C2.3** select suitable finishes (*e.g., stain, paint, varnish, oil, wax*), taking into account the type of material to be finished, the function of the finish, and the intended use of the project, and use appropriate methods to apply these finishes correctly (*e.g., brush, spray, roller*);
- C2.4** select and correctly install appropriate surface materials (*e.g., laminates, ceramics, butcher block, wood veneer*) on a variety of counters and/or furniture.

D. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of the environmental effects of the woodworking industry, and ways of reducing harmful effects;
- D2.** demonstrate an understanding of how the woodworking industry and society affect each other.

SPECIFIC EXPECTATIONS

D1. Technology and the Environment

By the end of this course, students will:

- D1.1** describe ways to improve air quality in a living or working space through the choice of materials for woodworking projects (*e.g., lumber, plastics, medium-density fibreboard, paint, varnish*);
- D1.2** assess the environmental effects of using scarce and/or exotic woods (*e.g., destruction of rainforest and old-growth boreal forests, displacement of Aboriginal peoples, loss of wildlife habitat*) and the extent to which sustainable forestry practices can reduce environmental degradation;
- D1.3** plan projects and apply strategies to minimize or mitigate degradation of the environment (*e.g., use efficient cutting patterns, reuse and recycle leftover materials, select sustainably produced products, contribute to restoration plans, purchase carbon offsets*);
- D1.4** assess various certifications and/or standards used to recognize sustainable practices (*e.g., Forest Stewardship Council Canada standards*).

D2. Technology and Society

By the end of this course, students will:

- D2.1** assess how consumer trends and technological innovations have affected employment in the custom woodworking industry (*e.g., use of exotic or old-growth woods, sale of prefabricated furniture and cabinets in large retail outlets, use of computer assisted design [CAD] and CNC machines*);
- D2.2** assess economic and societal issues related to the custom woodworking industry (*e.g., waste disposal, labour supply, logging near Aboriginal communities, imports and exports, use of renewable and non-renewable resources*).

E. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- E1.** explain and follow health and safety regulations and practices specific to woodworking;
- E2.** describe the knowledge and skills important for success in a custom woodworking business;
- E3.** demonstrate an understanding of careers in the custom woodworking industry, and the skills, education, and training required for these careers.

SPECIFIC EXPECTATIONS

E1. Health and Safety

By the end of this course, students will:

- E1.1** describe hazards related to woodworking materials, processes, tools, and equipment (*e.g., flammable or toxic fumes from solvents, paints, and varnishes; risk of injury from tools and equipment*);
- E1.2** describe an understanding of health and safety legislation, regulations, and standards that apply to the custom woodworking industry (*e.g., Workplace Hazardous Materials Information System [WHMIS], Workplace Safety and Insurance Board [WSIB] regulations, provincial labour legislation, Ontario Building Code, local by-laws*);
- E1.3** understand the rights and responsibilities of employees (*e.g., the right to know, the right to refuse, the right to participate, as outlined in the Occupational Health and Safety Act*);
- E1.4** demonstrate safe practices when using woodworking materials, processes, tools, equipment, and facilities;
- E1.5** demonstrate an understanding of how to select and use appropriate protective clothing, gear, and equipment (*e.g., dust mask, respirator, safety glasses, goggles, ventilation system*).

E2. Custom Woodworking Business

By the end of this course, students will:

- E2.1** describe the skills important for success as an entrepreneur in the custom woodworking industry (*e.g., communication, organization, creativity, problem solving*);

E2.2 describe the factors that should be considered when starting and/or operating a custom woodworking business (*e.g., potential market, products, competition, financing, business plan*);

E2.3 identify and compare the organization and management of unionized and non-unionized custom woodworking companies (*e.g., role of lead hand, shop steward, and contractor*);

E2.4 identify various types of business structures (*e.g., sole proprietorship, partnership, limited company*).

E3. Career Opportunities

By the end of this course, students will:

- E3.1** describe careers in the custom woodworking industry (*e.g., designer, trim carpenter, cabinet-maker, artisan, industrial woodworker, technician, technologist*);
- E3.2** describe the education and training required for careers in custom woodworking (*e.g., apprenticeship, postsecondary programs*);
- E3.3** explain the importance of lifelong learning for someone choosing a career in the custom woodworking industry;
- E3.4** demonstrate an understanding of and apply the Essential Skills that are important for success in the custom woodworking industry, as identified in the Ontario Skills Passport (*e.g., reading text, document use, measurement and calculation*);

E3.5 demonstrate an understanding of and apply the work habits that are important for success in the custom woodworking industry, as identified in the Ontario Skills Passport (*e.g., teamwork, customer service, entrepreneurship*);

E3.6 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in custom woodworking (*e.g., Passport to Safety certificate, technical drawings, reports, photographs of completed projects, Ontario Skills Passport Work Plan and Transition Plan*), and explain why having a current portfolio is important for career development and advancement.